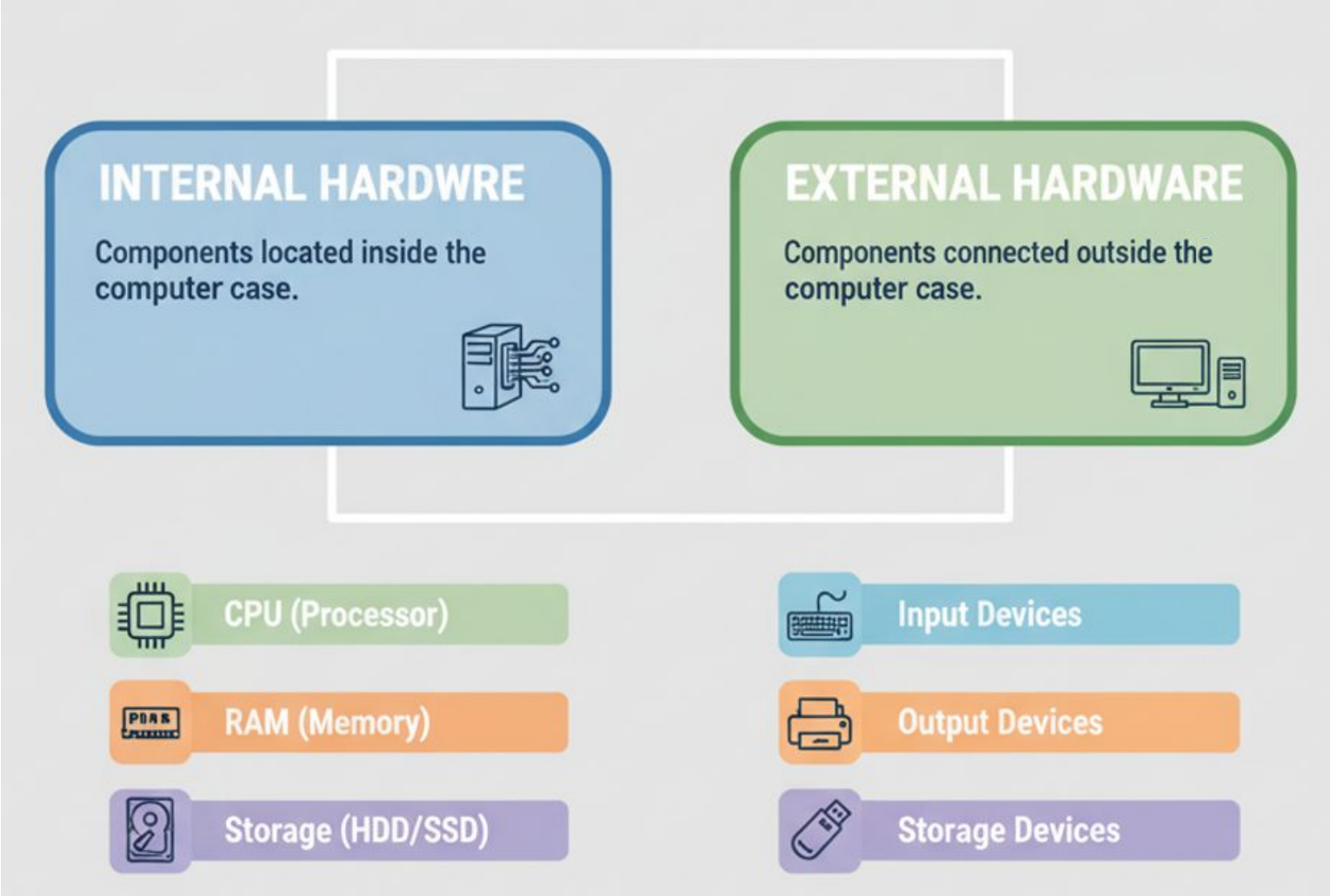
SWE 4101

Hardware Components & Peripherals

Lec - 2

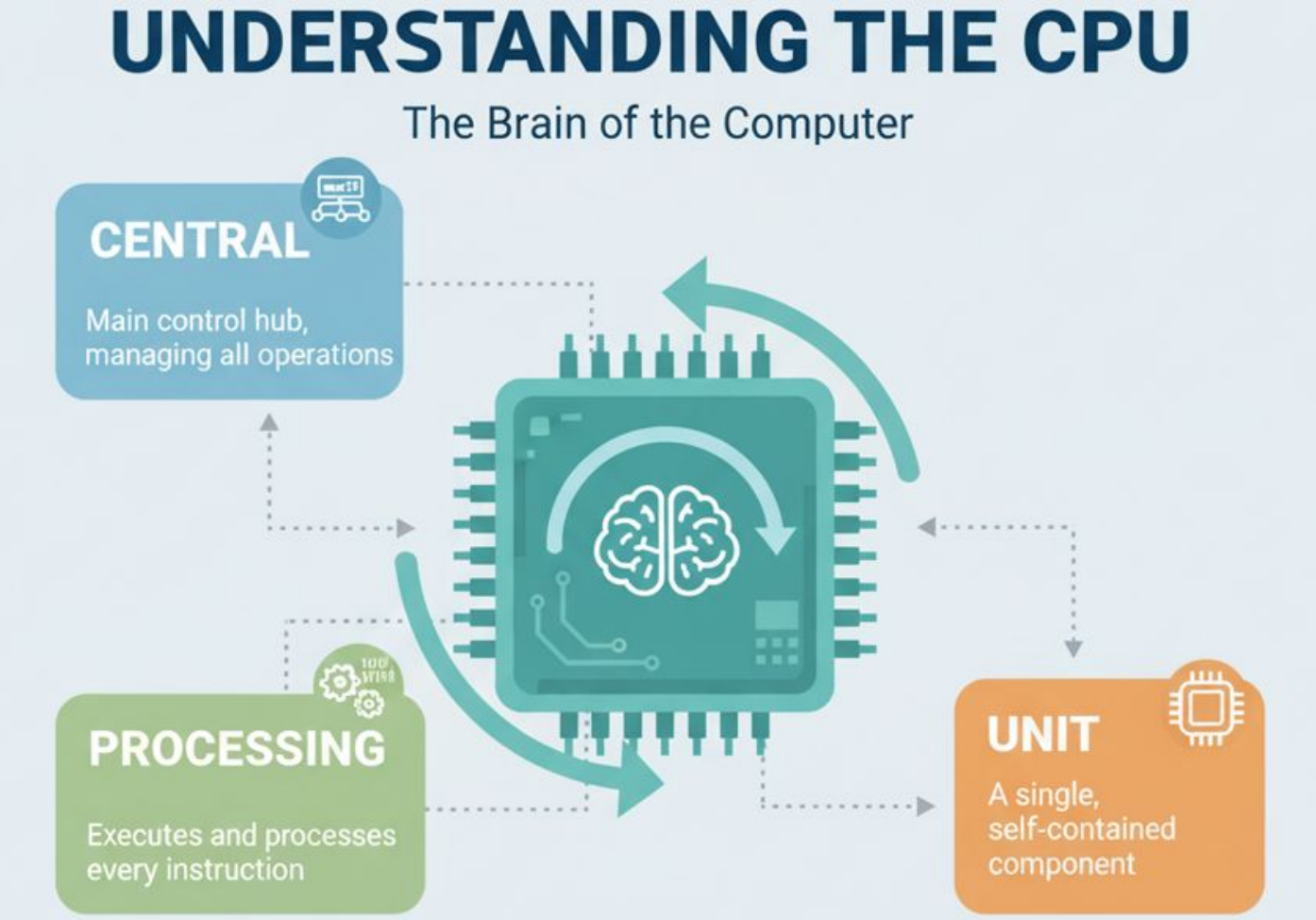
What is Computer Hardware

✔ Hardware is the **physical**, **tangible** parts of a computer system that you can **touch.**



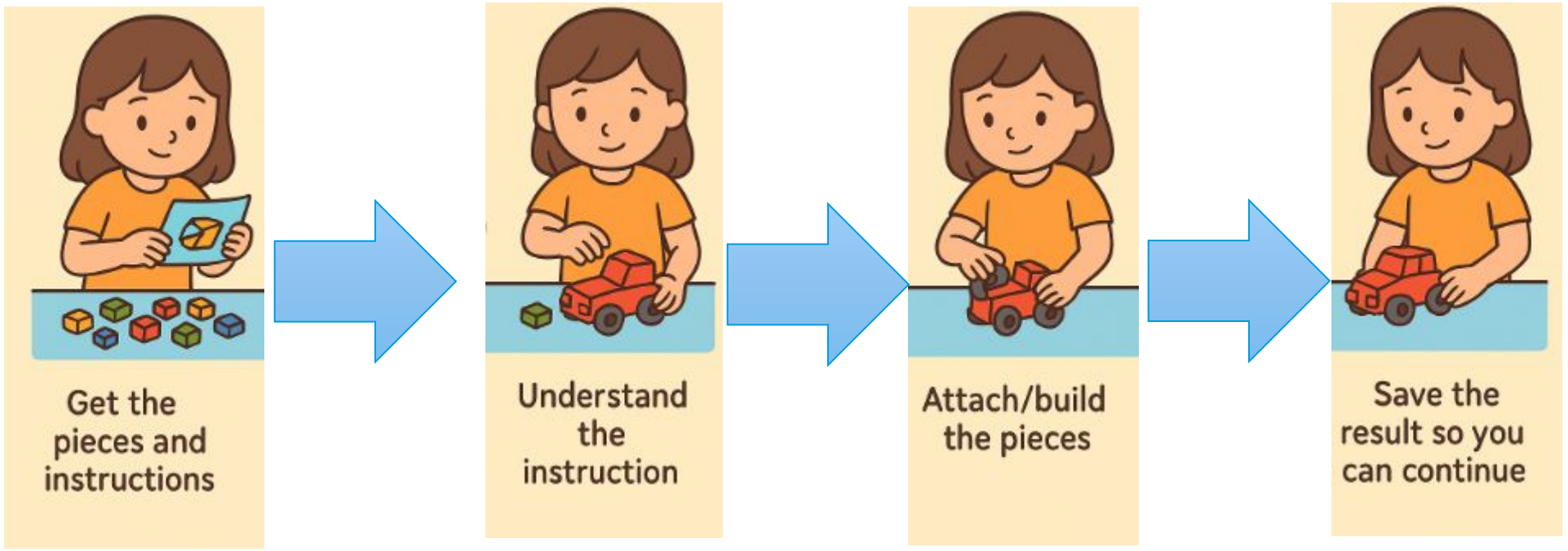
CPU (Central Processing Unit)

- ✔ **computer's brain**. The CPU was invented by Intel in 1971.
- ✔ handles all the tasks you tell the computer to do, like opening a game or writing a document.



CPU (Central Processing Unit)

✔ **How it works?**



FETCH

The CPU collects instructions from the system's memory.

DECODE

It changes those instructions into easy-to-follow steps.

EXECUTE

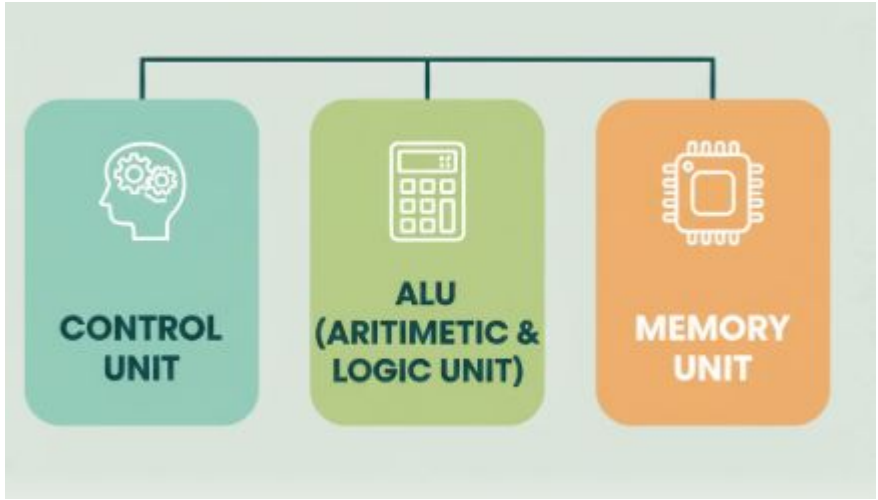
The CPU performs the action, such as solving math or moving information.

STORE

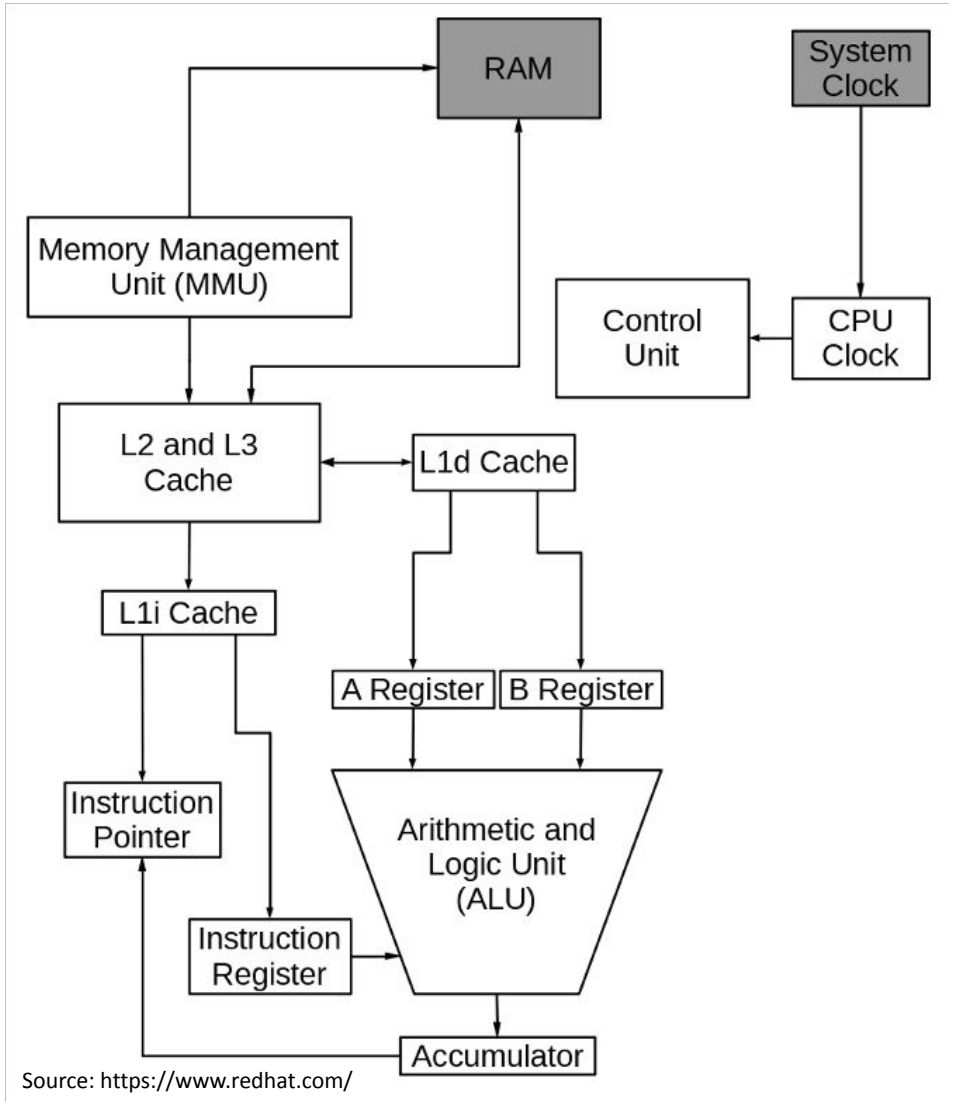
The output is kept in memory so it can be used later when required.

CPU (Central Processing Unit)

✔ Parts of CPU

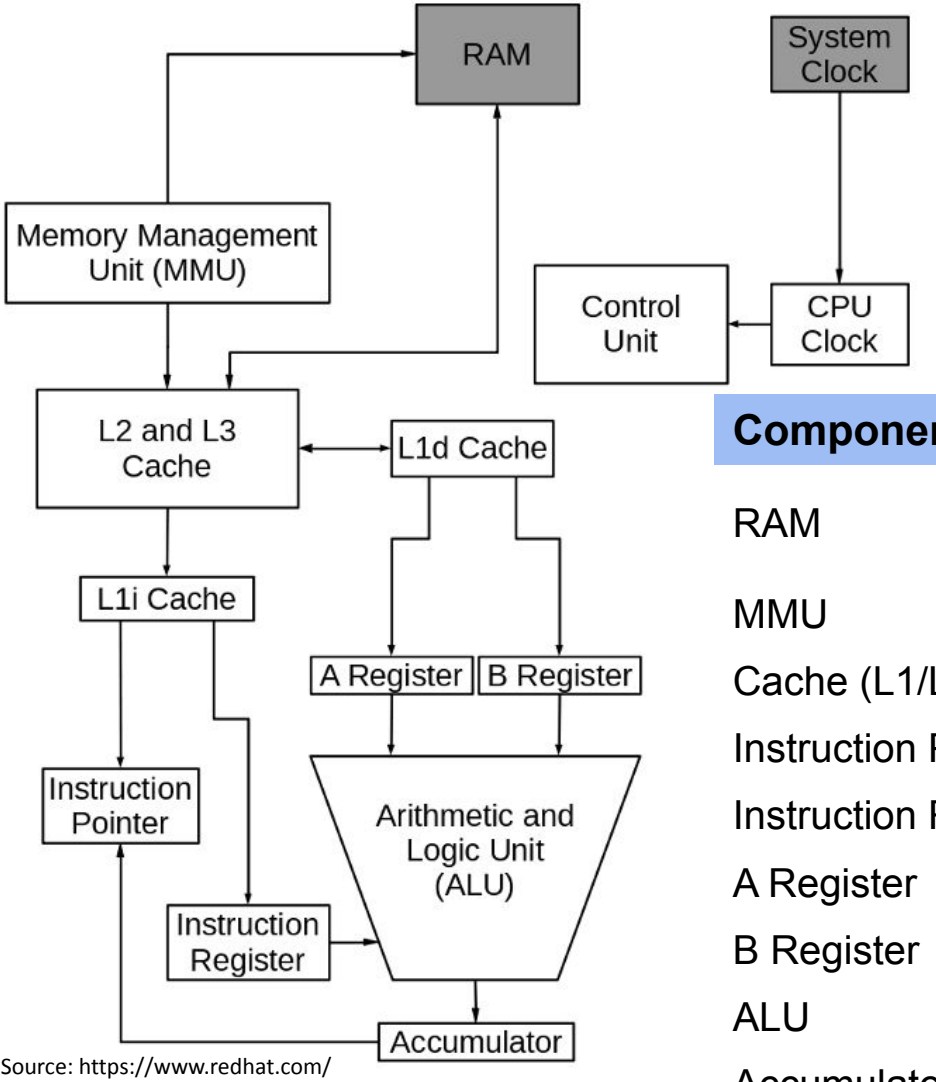


Control Unit (CU): Directs the flow of instructions and data within the computer.
Arithmetic Logic Unit (ALU): Performs calculations and logical operations.
Registers: Small, high-speed storage locations for temporarily holding data.
Cache Memory: Extremely fast memory that provides rapid access to frequently used data and instructions.
Buses: Pathways that carry data and signals between various parts of the computer.



Source: https://www.redhat.com/

Working Procedure



Source: https://www.redhat.com/

Let's add **two numbers** , **example** : 5 + 3.

Component	Action	Value / Example
RAM	Stores instructions and data	`LOAD A,5, LOAD B,3, ADD A,B`
MMU	Maps virtual → physical addresses	`0x01 → 0x101`
Cache (L1/L2/L3)	Holds frequently used instructions/data	`LOAD A,5`
Instruction Pointer	Points to next instruction	`0x01 → 0x02`
Instruction Register	Holds current instruction	`LOAD A,5`
A Register	Holds first operand	5
B Register	Holds second operand	3
ALU	Performs arithmetic/logic	5 + 3 = 8
Accumulator	Stores result	8

CPU (Central Processing Unit)

✔ Types of CPU

Based on purpose

General Purpose CPU : A CPU designed to handle a wide variety of tasks and applications, not limited to a specific function. For example Intel Core i7, AMD Ryzen 7.

Special Purpose CPU : A CPU designed for a specific task or application, optimized for that function rather than general use. GPU processors in graphics cards (like NVIDIA RTX) for rendering graphics, or microcontrollers in washing machines.

Based on Architecture

CISC (Complex Instruction Set Computer): A CPU architecture with a large set of instructions, allowing a single instruction to execute multiple low-level operations. Intel x86 processors — one instruction can move, add, and store data in a single command

RISC (Reduced Instruction Set Computer): A CPU architecture with a small set of simple instructions that execute very quickly, often requiring multiple instructions for complex tasks. ARM processors in smartphones (like Apple M1, Qualcomm Snapdragon).

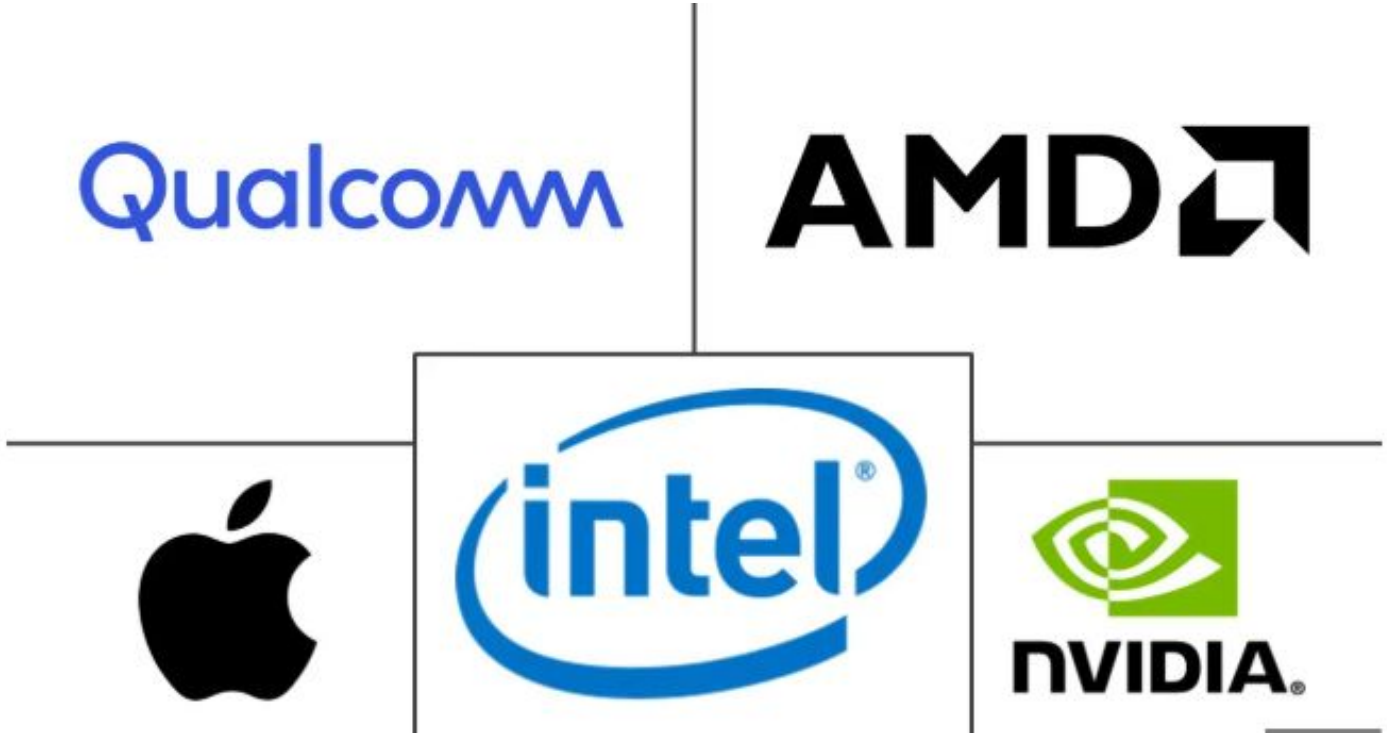
Based on Core

Single-Core CPU: A CPU with only one processing unit (core) that can execute one instruction at a time. Early Intel Pentium 4 processors.

Multi-Core CPU: A CPU with two or more cores, allowing it to execute multiple instructions simultaneously, improving multitasking and performance. Intel Core i9 (8-core/16-thread), AMD Ryzen 9 (12-core/24-thread).

CPU (Central Processing Unit)

- ✔ **Major CPU Manufacturers**



Memory Unit

- ✔ One of the most vital components of a computer system.
- ✔ Without memory, a computer would not be able to perform even the simplest tasks.

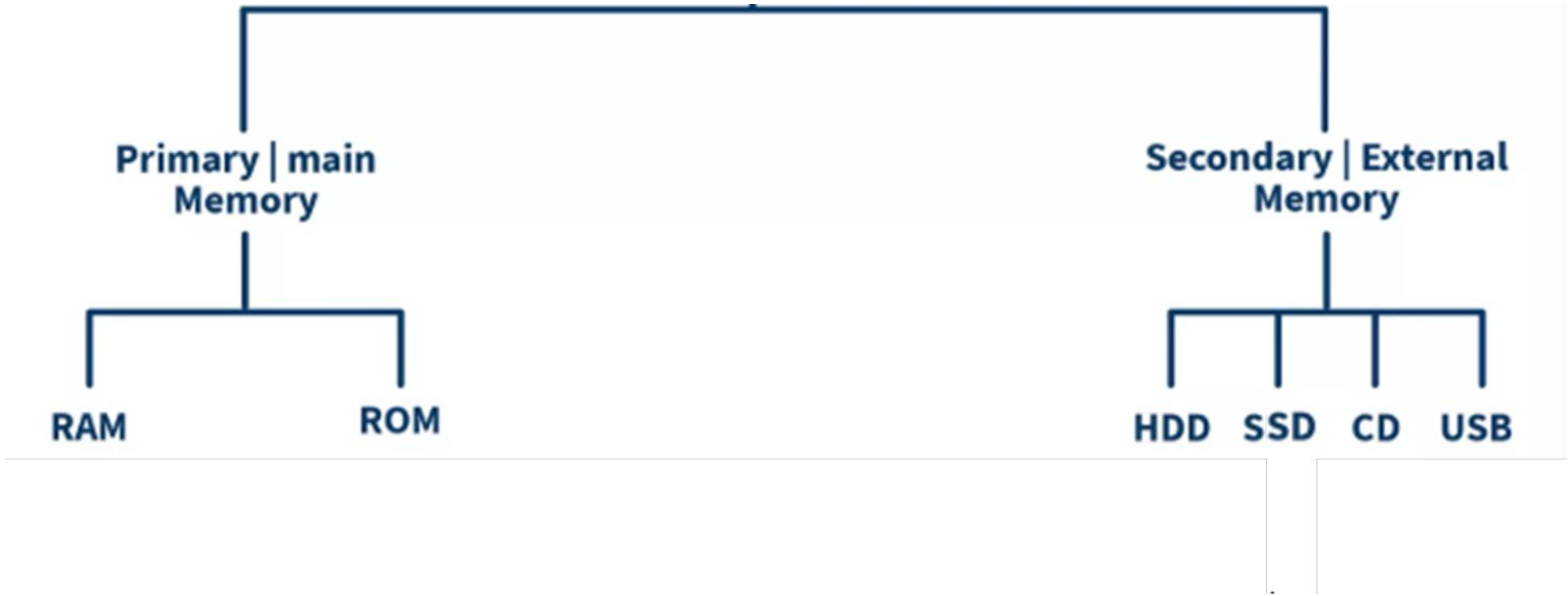
Understanding Memory Units

- ✔ The storage capacity of computer memory is measured in **bytes**.
- ✔ Each byte consists of **8 bits**, and each bit represents a binary value—either 0 or 1.
- ✔ The standard memory units are as follows:

Unit	Symbol	Equivalent
Bit	b	Smallest unit of data (0 or 1)
Byte	B	8 bits
Kilobyte	KB	1,024 bytes
Megabyte	MB	1,024 KB
Gigabyte	GB	1,024 MB
Terabyte	TB	1,024 GB
Petabyte	PB	1,024 TB
Exabyte	EB	1,024 PB

Memory Unit

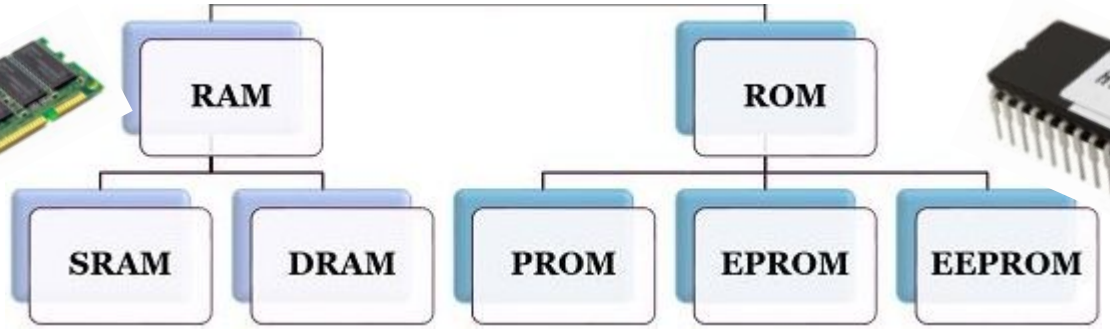
Types of Computer Memory



Memory Unit

Primary Memory







- ✔ **RAM** stands for Random Access Memory, and it stores data temporarily while the computer is in use.
- ✔ It helps the computer remember what you are doing at the moment.
- ✔ Volatile (loses data when powered off)

How does RAM work?

1. You turn on your computer, and important files load into RAM.
2. When you open an app, RAM stores it for quick access and sends data to the CPU for processing.
3. As you open more apps, RAM divides its space among them.
4. Once you close the app or shut down the computer, everything in RAM is erased.

Memory Unit

Primary Memory (RAM)

Static RAM (SRAM)

✔ SRAM is a high-speed type of memory that doesn't need constant refreshing.

- Stores data as long as power is supplied
- Operates very fast
- More expensive than other types of RAM
- Commonly used in small amounts, such as in cache memory

Dynamic RAM (DRAM)

✔ DRAM is the most widely used type of RAM that constantly refreshes its data.

- Holds data only for a short time
- Requires continuous refreshing
- Slower than SRAM
- Less expensive and used as the main memory in most computers

DDR RAM (Double Data Rate RAM)

✔ DDR RAM is an improved version of DRAM that operates at higher speed and efficiency.

Different versions of DDR RAM:

DDR1: Oldest and slowest version
DDR2: Faster and more efficient than DDR1
DDR3: Common in older computers and laptops
DDR4: Very fast and widely used in modern systems
DDR5: Latest and fastest version designed for advanced computers

Memory Unit

Primary Memory (RAM)

How Much RAM Do You Need?

- ✔ The amount of RAM you need depends on your usage.
- ✔ Having more RAM makes your computer run faster and more smoothly.



You have multiple browser tabs, a video editing software, and a music player running at the same time. Suddenly, your computer freezes and takes a long time to respond.

Why did this happen?

Memory Unit

Primary Memory (ROM)

- ✔ Read-only memory (ROM), is a type of **non-volatile memory** in a computer, which means the data stored in it **remains even when the power is turned off**.
- ✔ primarily used to store **permanent instructions** that the computer needs to start and operate, such as the BIOS or firmware.

PROM (Programmable ROM)

✔ A type of ROM that can be programmed **only once** after manufacturing.

- Once written, the data **cannot be changed**

Example: Used in electronic devices where permanent instructions are needed, like calculators or microwave controllers.

EPROM (Erasable Programmable ROM)

✔ A type of ROM that can be **erased and reprogrammed**.

- Data is erased using **ultraviolet (UV) light**
- Can be reprogrammed multiple times

Example: Used in BIOS chips of computers in earlier days

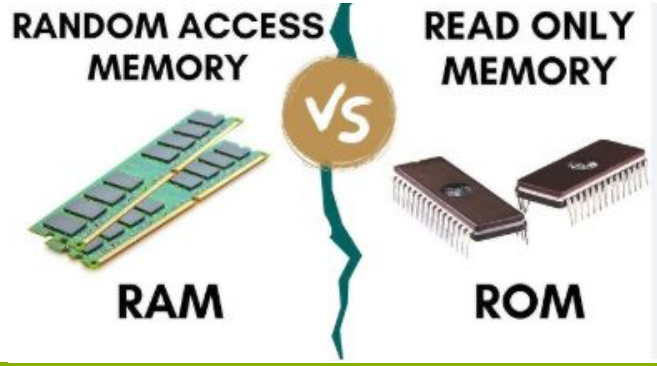
EEPROM (Electrically Erasable Programmable ROM)

✔ A type of ROM that can be **erased and reprogrammed electrically** without removing it from the system.

- Data is erased using **electrical signals**
- Can be rewritten multiple times

Example: Modern BIOS chips, memory cards, and embedded systems

RAM vs ROM



Feature	RAM (Random Access Memory)	ROM (Read-Only Memory)
Type of Memory	Volatile	Non-volatile
Data Loss	Data is lost when power is off	Data remains safe
Use	Temporary storage	Permanent instructions
Speed	Very fast	Slower than RAM
Editable	Can be changed	Cannot be easily changed
Storage Capacity	High	Low
Cost	Expensive	Cheaper
Working Stage	Active during processing	Active during booting
Example	DDR4, DDR5	BIOS, Firmware

Memory Unit

Secondary Memory

✔ **Secondary storage** refers to non-volatile storage devices that hold data permanently.

✔ keeps data for a long time without the need for a continuous power source.

Importance of Storage Devices

- Provide permanent storage for data
- Allow storage of large volumes of information
- Serve as a backup solution
- Enable data recovery in case of system failures
- Facilitate file sharing and data transfer



Categories of Secondary Storage Devices

Category	Examples	Speed	Portability	Use Case
Magnetic Storage	HDD, Tape Drives	Moderate	Low	Long-term storage
Solid-State Storage	SSD, USB Drives	High	High	Fast data access
Optical Storage	CDs, DVDs, Blu-rays	Low	Moderate	Media storage
Cloud Storage	Google Drive, Dropbox	High	Very High	Remote backups

Memory Unit

Secondary Memory (HDD)

✔ A hard disk drive is used to store a large amount of data.
✔ HDD is a **type of magnetic storage device**.

Key Features of HDD:

- Stores large amounts of data (500 GB, 1 TB, or more)
- Non-volatile; retains data without power
- Contains moving parts (spinning disks)
- Uses magnetic storage technology
- Installed inside the computer case
- Slower than SSD
- Low-cost storage option



Memory Unit

Secondary Memory (SSD)

✔ A solid-state drive stores data in a fast way.
✔ It has no moving parts. It works silently and quickly.

Key Features of SSD:

- Stores large data (256 GB, 512 GB, 1 TB)
- Non-volatile; retains data without power
- Uses flash memory for storage
- Faster than HDD
- Energy-efficient; saves laptop battery
- Lightweight and compact
- Silent operation (no moving parts)



Memory Unit

Secondary Memory (Magnetic Tape)

- ✔ A magnetic tape stores data on a long plastic strip.
- ✔ It looks like the tape inside a cassette.
- ✔ Can save a lot of data for a long time. slow but cheap.
- ✔ Industries use it to keep backup files.

Key Features of Magnetic Tape:

- Stores very large data (multiple terabytes)
- Non-volatile; retains data without power
- Sequential data storage (one after another)
- Slow read/write speed
- Low-cost for long-term storage
- Portable and easy to carry
- Requires a tape drive to read/write
- Type of magnetic storage device



Memory Unit

Secondary Memory (Others)



Floppy Disk:

- ✔ Stores 1.44 MB, magnetic storage
- ✔ Lightweight and portable
- ✔ Slow read/write, requires floppy drive
- ✔ Obsolete; replaced by better devices



SD Card:

- ✔ Stores data from a few MB to 1 TB+
- ✔ Non-volatile; retains data without power
- ✔ No moving parts; silent operation
- ✔ Small and lightweight
- ✔ Used in phones, cameras, tablets
- ✔ Removable and reusable
- ✔ Faster read/write than CDs/DVDs



USB Flash Drive:

- ✔ Stores data from 2 GB to 1 TB+
- ✔ No moving parts; durable
- ✔ Uses solid-state storage
- ✔ Fast data transfer
- ✔ Reusable; can delete and save data multiple times
- ✔ Connects via USB



Compact Disc (CD):

- ✔ Stores up to 700 MB, non-volatile
- ✔ Uses laser (optical) storage
- ✔ Lightweight and portable
- ✔ Requires CD drive
- ✔ Low-cost storage option



DVD:

- ✔ Stores 4.7–8.5 GB, non-volatile
- ✔ Optical storage for videos, software, files
- ✔ Lightweight and portable
- ✔ Requires DVD drive
- ✔ Rewritable DVDs (DVD-RW) are reusable and cheap



Blu-ray Disc:

- ✔ Stores 25 GB (single layer) or 50 GB (dual layer)
- ✔ Non-volatile; retains data without power
- ✔ Uses blue laser for reading/writing
- ✔ Better picture and sound than DVD
- ✔ Lightweight, portable, and scratch-resistant
- ✔ Works only in Blu-ray players or drives

Memory Unit

Cache Memory

- ✔ Cache is the temporary memory officially termed “CPU cache memory.”
- ✔ is a high-speed memory location built into or very close to a computer’s processor.
- ✔ stores copies of frequently used data and instructions so that the CPU can access them much more quickly than if it had to fetch them from main memory (RAM) or from disk.

Why Cache Matters (Performance Impact)

- ✔ When the CPU needs to access data, it checks the cache first.
- ✔ If the data is found (“cache hit”), the access is extremely fast.
- ✔ If it’s not found (“cache miss”), the system must fetch it from slower memory or storage which delays performance.

Levels of Cache

Modern CPUs typically implement multiple cache levels to balance speed and size:

- ✔ **L1 Cache**: The smallest and fastest—built directly into the CPU. It holds the data most likely to be used next.
- ✔ **L2 Cache**: A larger but slightly slower cache that the CPU checks if data isn’t found in L1.
- ✔ **L3 and beyond / Other caches**: Some systems include L3 or even L4. There is also disk cache (on hard drives/SSDs) and GPU cache in graphics systems.

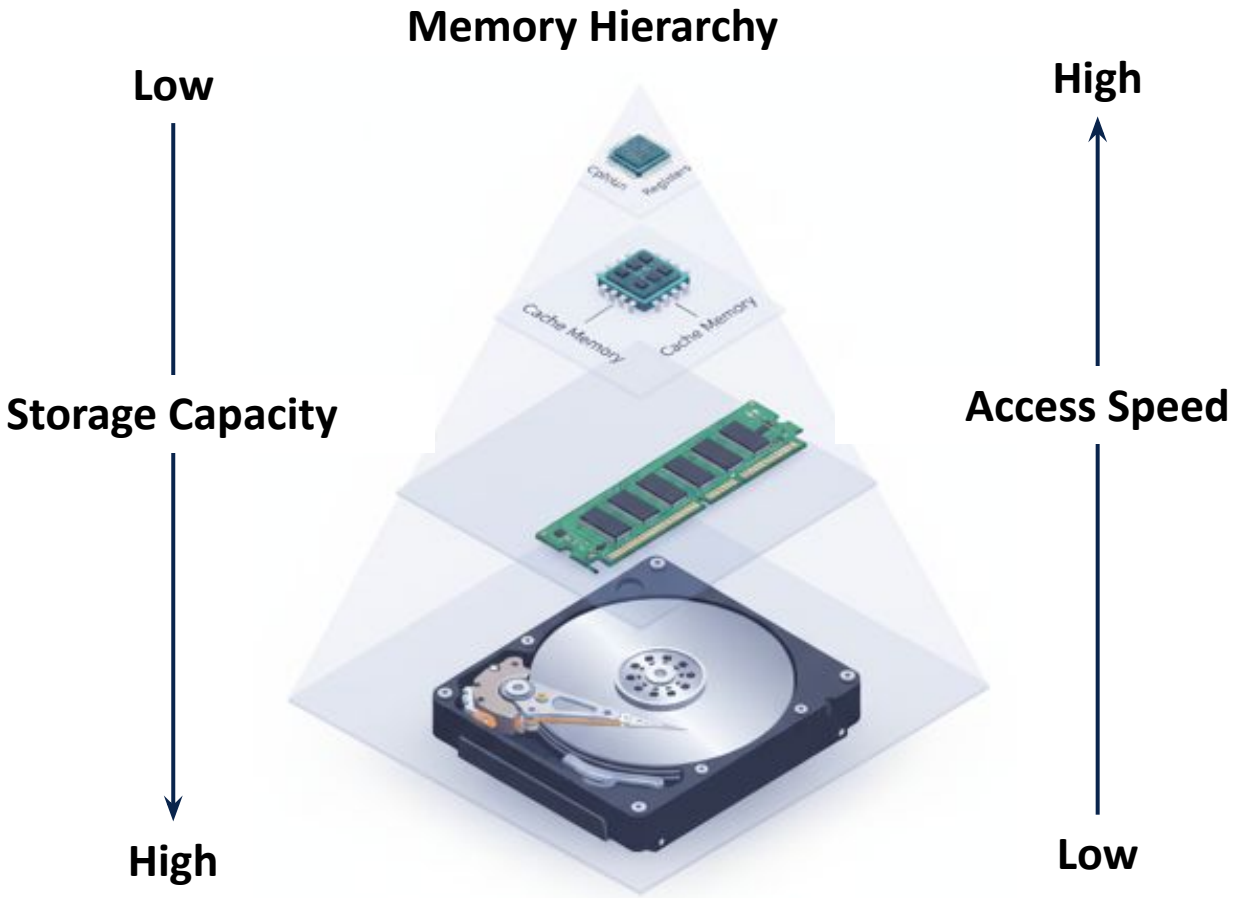
When you’re buying or evaluating a computer, cache is one of many specs—but not always the most visible or advertised.

- ✔ **Cache size**: A larger cache can hold more data, but may also have higher latency (slower access).
- ✔ **Latency/speed**: The access speed of the cache matters more than size.

Memory Unit

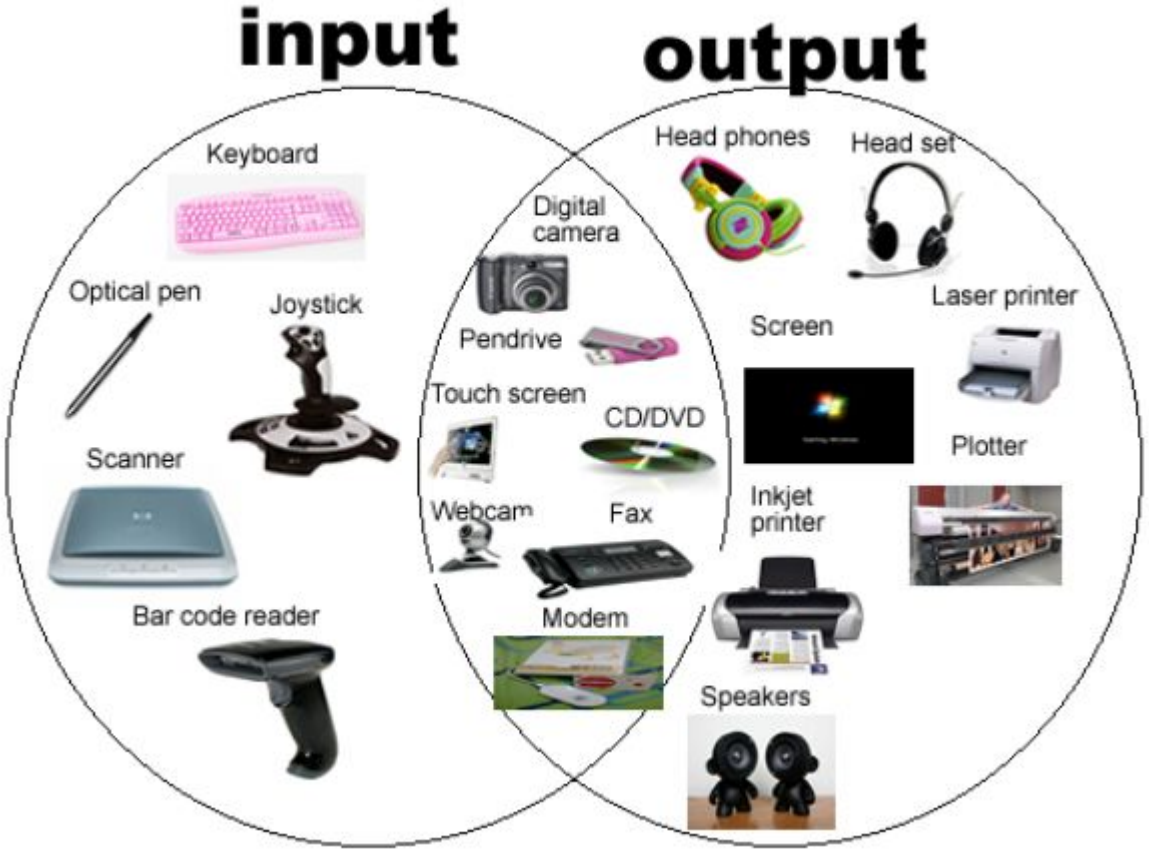
Register

- ✔ are a type of computer memory built directly into the processor or CPU (Central Processing Unit)
- ✔ is used to store and manipulate data during the execution of instructions.
- ✔ stores copies of frequently used data and instructions so that the CPU can access them much more quickly than if it had to fetch them from main memory (RAM) or from disk.



Input/Output Devices

Input Device: Hardware used to enter data into a computer, e.g., keyboard, mouse.
Output Device: Hardware that displays or produces the result of processed data, e.g., monitor, printer.



Assignment: Input and Output Devices

Objective:

Understand the different types of input and output devices, their functions, working principles, and importance in computer systems.

Instructions:

1. List at least **5 input devices** and **5 output devices**.
2. For each device, provide the following details:
    1. **Device Name**
    2. **Type** (Input/Output)
    3. **Function / Purpose**
    4. **How it works** (brief explanation)
    5. **Example Usage** (where it is commonly used)
    6. **Additional Info** (optional, e.g., advantages, limitations, special features)

Submission deadline: